

Well-Posedness of the Linear Stationary Micropolar System on the Divergence-Free Space

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Abstract

In Lesson 1 I worked through three scalar elliptic problems and showed that boundedness and coercivity of the bilinear form are what give well-posedness. These notes carry the same programme through for a coupled system, namely the two dimensional stationary linear micropolar equations, restricted to divergence free velocity fields so that the pressure disappears from the formulation. The new difficulty is that the coupling terms have no sign, so they can work against coercivity, and the tool for dealing with them is Young's inequality with a free parameter. The main result is that coercivity holds with a constant that does not involve the vortex viscosity at all, while boundedness gives a constant that grows linearly in it. I explain at the end what that difference means for the numerical behaviour of the system, and why it is what one should expect physically.

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1 Results we will use

Everything in these notes is built from the results collected here. I have stated them all in one place because the same few are used again and again, and it is easier to follow the argument if it is clear at every step which one is being applied. The first three are from Lesson 1. Lemmas 1.4, 1.6 and 1.7 are new.

Throughout, $\Omega \subset \mathbb{R}^2$ is a bounded Lipschitz domain with boundary $\partial\Omega$, all derivatives are weak derivatives, and I write $\partial_1 = \partial/\partial x$ and $\partial_2 = \partial/\partial y$. The L^2 inner product is

$$(\phi, \psi) = \int_{\Omega} \phi \psi \, dx \quad \text{or} \quad (\phi, \psi) = \int_{\Omega} \phi \cdot \psi \, dx$$

depending on whether the arguments are scalars or vectors.

Lemma 1.1 (Cauchy–Schwarz). *For $\phi, \psi \in L^2(\Omega)$,*

$$|(\phi, \psi)| \leq \|\phi\|_{L^2} \|\psi\|_{L^2}.$$

Lemma 1.2 (Poincaré–Friedrichs). *There is a constant $C_{\Omega} > 0$ depending only on Ω such that*

$$\|\phi\|_{L^2} \leq C_{\Omega} \|\nabla \phi\|_{L^2} \quad \text{for every } \phi \in H_0^1(\Omega).$$

For a vector field $\mathbf{v} \in [H_0^1(\Omega)]^2$ the same bound holds with the same constant, by applying it to each component and adding.

Lemma 1.3 (Lax–Milgram). *Let V be a Hilbert space and let $a : V \times V \rightarrow \mathbb{R}$ be bilinear with*

$$|a(u, v)| \leq M \|u\|_V \|v\|_V \quad \text{and} \quad a(u, u) \geq \alpha \|u\|_V^2$$

for some constants $M, \alpha > 0$. Let $F \in V'$ be a bounded linear functional. Then there is exactly one $u \in V$ with $a(u, v) = F(v)$ for all $v \in V$, and it satisfies $\|u\|_V \leq \alpha^{-1} \|F\|_{V'}$.

Lemma 1.4 (Young’s inequality with a free parameter). *For $a, b \in \mathbb{R}$ and any $\varepsilon > 0$,*

$$|ab| \leq \frac{\varepsilon}{2} a^2 + \frac{1}{2\varepsilon} b^2.$$

Proof. Since squares are non-negative,

$$\left(\sqrt{\varepsilon} a - \frac{b}{\sqrt{\varepsilon}} \right)^2 \geq 0.$$

Expanding the square,

$$\varepsilon a^2 - 2ab + \frac{b^2}{\varepsilon} \geq 0,$$

so $2ab \leq \varepsilon a^2 + \varepsilon^{-1} b^2$, and dividing by 2 gives the result for ab . Replacing b by $-b$ gives the same bound for $-ab$, so the inequality holds for $|ab|$. $\square$

Remark 1.5 (What the free parameter is for). This is the tool I will use to handle the coupling terms, so it is worth being clear about what it does. Suppose an estimate produces a cross term ab that we cannot control directly, but we do have squared terms a^2 and b^2 available with known positive coefficients. Young's inequality converts the cross term into two squares, and the parameter ε decides how the cost is split between them. Taking ε small puts most of the cost on b^2 and very little on a^2 . Taking ε large does the opposite. So we choose ε so that each piece is small enough to be swallowed by a term we already have. This is what is meant by absorbing a term, and it is the single most used move in this subject.

For the next two results I use the following formula, which is just one dimensional integration by parts applied in the x_i direction. For scalar functions ϕ, ψ and either index $i \in \{1, 2\}$,

$$\int_{\Omega} (\partial_i \phi) \psi \, dx = \int_{\partial\Omega} \phi \psi n_i \, ds - \int_{\Omega} \phi (\partial_i \psi) \, dx, \quad (1)$$

where n_i is the i th component of the outward unit normal. If either ϕ or ψ vanishes on $\partial\Omega$ the boundary term disappears and moving a derivative from one factor to the other costs a minus sign. Every identity below comes from (1) applied one index at a time.

In two dimensions we use the operators

$$\begin{aligned} \nabla w &= (\partial_1 w, \partial_2 w), & \operatorname{div} \mathbf{v} &= \partial_1 v_1 + \partial_2 v_2, \\ \operatorname{curl} \mathbf{v} &= \partial_1 v_2 - \partial_2 v_1, & \nabla^\perp w &= (\partial_2 w, -\partial_1 w). \end{aligned}$$

The last two are special to two dimensions. In three dimensions the curl of a vector is a vector, but in two dimensions it has only one non-zero component so we treat it as a scalar. The operator $\nabla^\perp$ is the gradient rotated by ninety degrees, and it turns a scalar into a vector. These two operators are how the micropolar system converts between the vector velocity and the scalar microrotation.

Lemma 1.6 (Rotation identity). *For $w \in H_0^1(\Omega)$ and $\mathbf{v} \in [H_0^1(\Omega)]^2$,*

$$(\nabla^\perp w, \mathbf{v}) = (w, \operatorname{curl} \mathbf{v}).$$

Proof. Expanding the dot product into components,

$$(\nabla^\perp w, \mathbf{v}) = \int_{\Omega} \nabla^\perp w \cdot \mathbf{v} \, dx = \int_{\Omega} (\partial_2 w) v_1 \, dx - \int_{\Omega} (\partial_1 w) v_2 \, dx.$$

Apply (1) to each term with $\phi = w$, moving the derivative off w and onto the component of $\mathbf{v}$. The boundary terms vanish because v_1 and v_2 have zero trace. This gives

$$\int_{\Omega} (\partial_2 w) v_1 \, dx = - \int_{\Omega} w (\partial_2 v_1) \, dx, \quad - \int_{\Omega} (\partial_1 w) v_2 \, dx = + \int_{\Omega} w (\partial_1 v_2) \, dx.$$

Adding,

$$(\nabla^\perp w, \mathbf{v}) = \int_{\Omega} w (\partial_1 v_2 - \partial_2 v_1) \, dx = \int_{\Omega} w \operatorname{curl} \mathbf{v} \, dx = (w, \operatorname{curl} \mathbf{v}). \quad \square$$

Lemma 1.7 (Curl and divergence decomposition). *For $\mathbf{v} \in [H_0^1(\Omega)]^2$,*

$$\|\nabla \mathbf{v}\|_{L^2}^2 = \|\operatorname{curl} \mathbf{v}\|_{L^2}^2 + \|\operatorname{div} \mathbf{v}\|_{L^2}^2,$$

and consequently

$$\|\operatorname{curl} \mathbf{v}\|_{L^2} \leq \|\nabla \mathbf{v}\|_{L^2}.$$

Proof. Write out all three quantities in components:

$$\begin{aligned}\|\nabla \mathbf{v}\|_{L^2}^2 &= \int_{\Omega} [(\partial_1 v_1)^2 + (\partial_2 v_1)^2 + (\partial_1 v_2)^2 + (\partial_2 v_2)^2] \, dx, \\ \|\operatorname{curl} \mathbf{v}\|_{L^2}^2 &= \int_{\Omega} [(\partial_1 v_2)^2 - 2(\partial_1 v_2)(\partial_2 v_1) + (\partial_2 v_1)^2] \, dx, \\ \|\operatorname{div} \mathbf{v}\|_{L^2}^2 &= \int_{\Omega} [(\partial_1 v_1)^2 + 2(\partial_1 v_1)(\partial_2 v_2) + (\partial_2 v_2)^2] \, dx.\end{aligned}$$

Adding the last two, the four squared terms are exactly those appearing in $\|\nabla \mathbf{v}\|_{L^2}^2$, and what is left over is

$$2 \int_{\Omega} (\partial_1 v_1)(\partial_2 v_2) \, dx - 2 \int_{\Omega} (\partial_1 v_2)(\partial_2 v_1) \, dx. \quad (2)$$

So the claim will follow once we show that the two integrals in (2) are equal.

Start from the first one and apply (1) with $\phi = v_2$, $\psi = \partial_1 v_1$ and $i = 2$, that is, move ∂_2 off v_2 . The boundary term vanishes because v_2 has zero trace, so

$$\int_{\Omega} (\partial_1 v_1)(\partial_2 v_2) \, dx = - \int_{\Omega} (\partial_2 \partial_1 v_1) v_2 \, dx.$$

Since mixed partial derivatives commute we may write $\partial_2 \partial_1 v_1 = \partial_1(\partial_2 v_1)$, so the integrand again has the form $(\partial_1 \phi)\psi$ with $\phi = \partial_2 v_1$ and $\psi = v_2$. Applying (1) a second time, now with $i = 1$, and noting the boundary term vanishes for the same reason,

$$- \int_{\Omega} \partial_1(\partial_2 v_1) v_2 \, dx = + \int_{\Omega} (\partial_2 v_1)(\partial_1 v_2) \, dx.$$

Chaining the two steps,

$$\int_{\Omega} (\partial_1 v_1)(\partial_2 v_2) \, dx = \int_{\Omega} (\partial_2 v_1)(\partial_1 v_2) \, dx,$$

so (2) is zero and the identity follows. Two applications of (1) produce two minus signs, which cancel, and that is the whole proof.

For the inequality, $\|\operatorname{div} \mathbf{v}\|_{L^2}^2 \geq 0$ because it is a squared norm, so dropping it from the right hand side can only make that side smaller:

$$\|\nabla \mathbf{v}\|_{L^2}^2 \geq \|\operatorname{curl} \mathbf{v}\|_{L^2}^2,$$

and taking square roots gives the claim. $\square$

Remark 1.8. The move at the end, discarding a non-negative summand to get an inequality, is the same one used in Lesson 1 to obtain $\|\nabla \phi\|_{L^2} \leq \|\phi\|_{H^1}$ from $\|\phi\|_{H^1}^2 = \|\phi\|_{L^2}^2 + \|\nabla \phi\|_{L^2}^2$. It comes up constantly and is worth recognising on sight.

2 The equations and what the terms mean

2.1 The model

The micropolar model, due to Eringen, describes a fluid whose suspended microstructure carries its own rotational degrees of freedom. Ordinary Navier–Stokes treats fluid particles as points with a velocity and nothing else. The micropolar model gives them a spin as well, and tracks it as a second field. It is used for blood, where the red cells rotate, for ferrofluids, for liquid crystals, and for lubricants with suspended additives.

In two dimensions, in the stationary linear case, the system is

$$-(\nu + \nu_r)\Delta \mathbf{u} + \nabla p = 2\nu_r \nabla^\perp w + \mathbf{f}, \quad (3)$$

$$\operatorname{div} \mathbf{u} = 0, \quad (4)$$

$$-(c_a + c_d)\Delta w + 4\nu_r w = 2\nu_r \operatorname{curl} \mathbf{u} + g, \quad (5)$$

with $\mathbf{u} = \mathbf{0}$ and $w = 0$ on $\partial\Omega$. The unknowns are the velocity $\mathbf{u}$, the pressure p , and the microrotation w , which is the angular velocity of the microstructure.

2.2 What each term is doing

It is worth going through the terms one at a time, because the whole point of the analysis is to find out how they interact, and that is much easier to follow if their meanings are already clear.

- $-(\nu + \nu_r)\Delta \mathbf{u}$ is viscous diffusion of momentum. The coefficient is the sum of two viscosities. The first, ν , is the ordinary dynamic viscosity, the same one as in Navier–Stokes. The second, ν_r , is called the vortex viscosity, and it is the extra resistance the fluid feels because the microstructure is there at all. Note that this term is where ν_r first appears, and it appears with a plus sign, so at this stage it looks like it can only help.
- ∇p is the pressure gradient. Pressure enters the momentum equation only through its gradient, which will matter shortly.
- $2\nu_r \nabla^\perp w$ is the first coupling term. It says that where the microstructure is spinning, that spin drags the bulk fluid along. This is a transfer of angular momentum from the microrotation field to the velocity field.
- $\operatorname{div} \mathbf{u} = 0$ is incompressibility.
- $-(c_a + c_d)\Delta w$ is angular diffusion. Spin spreads out through the microstructure in the same way that momentum spreads out through the fluid. The constants c_a and c_d are angular viscosities, and they only ever appear in the combination $c_a + c_d$ in the two dimensional case, so I will treat that sum as a single positive constant.
- $4\nu_r w$ is a damping or reaction term. It says that spin decays. If the microstructure spins faster than the surrounding fluid rotates, friction slows it down. It is the only zeroth order term in the whole system, and it will turn out to be very important.
- $2\nu_r \operatorname{curl} \mathbf{u}$ is the second coupling term, the mirror image of the first. Where the bulk fluid is rotating, it spins up the microstructure. This is the transfer in the opposite direction.

The key observation, which is the theme of these notes, is that ν_r appears in four places: in the velocity diffusion, in the damping term, and in both couplings. It is not four independent parameters. It is one physical quantity, the vortex viscosity, appearing four times, and the coefficients $2\nu_r$, $2\nu_r$ and $4\nu_r$ are related to each other, not free. That relationship is what makes the analysis below work out as cleanly as it does.

2.3 Why we work on the divergence-free space

To keep the analysis to one difficulty at a time, I work here on the space of divergence free velocity fields,

$$\mathbf{V}_{\operatorname{div}} = \{ \mathbf{v} \in [H_0^1(\Omega)]^2 : \operatorname{div} \mathbf{v} = 0 \}, \quad W = H_0^1(\Omega).$$

The reason is that when we derive the weak form, the pressure appears only in the term

$$-\int_{\Omega} p \operatorname{div} \mathbf{v} \, dx,$$

and if $\mathbf{v}$ is divergence free this vanishes identically. So the pressure drops out of the formulation completely, and we are left with a problem in two unknowns rather than three, on a single product space, which is exactly the setting Lax–Milgram is stated for.

This is legitimate mathematics and it gives us a genuine well-posedness result. It is not, however, how one discretises the problem, because building a finite element basis for $\mathbf{V}_{\text{div}}$ is hard. That difficulty is what forces the mixed formulation and the inf-sup condition, and it is the subject of the next set of notes. Doing it this way round means we get the coercivity result cleanly here, and meet the pressure as a separate problem rather than fighting both at once.

2.4 The product space

On $\mathbf{V}_{\text{div}} \times W$ we use the norm

$$\|(\mathbf{v}, z)\|^2 = \|\mathbf{v}\|_{H^1}^2 + \|z\|_{H^1}^2 = \|\mathbf{v}\|_{L^2}^2 + \|\nabla \mathbf{v}\|_{L^2}^2 + \|z\|_{L^2}^2 + \|\nabla z\|_{L^2}^2. \quad (6)$$

The idea is simply to treat the pair $(\mathbf{u}, w)$ as one unknown living in one space, so that everything from Lesson 1 applies with no structural change. This is the standard way to handle coupled systems and it is worth getting used to.

2.5 From three equations to one

The system (3) to (5) has three equations and three unknowns, but the analysis below is carried out on a single equation with a single bilinear form. It is worth doing that reduction slowly, because three separate things happen and it is easy to lose track of which is which. The incompressibility equation is absorbed into the choice of space, the pressure term then disappears as a consequence, and the two remaining equations are added together into one.

Step 1: the incompressibility equation becomes the space. The equation

$$\text{div } \mathbf{u} = 0$$

is a constraint on $\mathbf{u}$ rather than a differential equation we need to test against. So instead of multiplying it by a test function, we build it into the space in which we look for $\mathbf{u}$:

$$\mathbf{V}_{\text{div}} = \{\mathbf{v} \in [H_0^1(\Omega)]^2 : \text{div } \mathbf{v} = 0\}.$$

Searching for $\mathbf{u}$ in this space imposes $\text{div } \mathbf{u} = 0$ exactly. That disposes of one of the three equations, and two remain.

Step 2: the momentum equation. Multiply (3) by a test function $\mathbf{v} \in \mathbf{V}_{\text{div}}$ and integrate over Ω :

$$-(\nu + \nu_r) \int_{\Omega} (\Delta \mathbf{u}) \cdot \mathbf{v} \, dx + \int_{\Omega} \nabla p \cdot \mathbf{v} \, dx = 2\nu_r \int_{\Omega} \nabla^\perp w \cdot \mathbf{v} \, dx + \int_{\Omega} \mathbf{f} \cdot \mathbf{v} \, dx.$$

For the Laplacian term, apply Green's identity componentwise exactly as in Lesson 1. The boundary terms vanish because $\mathbf{v}$ has zero trace, giving

$$-(\nu + \nu_r) \int_{\Omega} (\Delta \mathbf{u}) \cdot \mathbf{v} \, dx = (\nu + \nu_r) (\nabla \mathbf{u}, \nabla \mathbf{v}).$$

For the pressure term, apply the master formula (1) with $\phi = p$ and $\psi = v_i$, once for each component. The boundary terms again vanish because $\mathbf{v}$ has zero trace, and each move costs a minus sign:

$$\int_{\Omega} \nabla p \cdot \mathbf{v} \, dx = \sum_{i=1}^2 \int_{\Omega} (\partial_i p) v_i \, dx = - \sum_{i=1}^2 \int_{\Omega} p (\partial_i v_i) \, dx = - \int_{\Omega} p \, \text{div } \mathbf{v} \, dx.$$

But $\mathbf{v} \in \mathbf{V}_{\text{div}}$, so $\text{div } \mathbf{v} = 0$ and the entire pressure term is zero.

This is the payoff from Step 1, and it is the whole reason for restricting the space. It is worth noticing that the pressure vanished because of the divergence of the *test* function, not of the solution. The weak form of the momentum equation is therefore

$$(\nu + \nu_r)(\nabla \mathbf{u}, \nabla \mathbf{v}) - 2\nu_r(\nabla^\perp w, \mathbf{v}) = (\mathbf{f}, \mathbf{v}) \quad \text{for all } \mathbf{v} \in \mathbf{V}_{\text{div}}. \quad (7)$$

Step 3: the microrotation equation. Multiply (5) by a test function $z \in W = H_0^1(\Omega)$ and integrate over Ω . Green's identity applies to the Laplacian term, with the boundary term vanishing because z has zero trace, and the zeroth order term $4\nu_r w$ contains no derivative so nothing needs to be done to it. This gives

$$(c_a + c_d)(\nabla w, \nabla z) + 4\nu_r(w, z) - 2\nu_r(\text{curl } \mathbf{u}, z) = (g, z) \quad \text{for all } z \in W. \quad (8)$$

Step 4: adding the two equations. We now have two weak equations, (7) tested against every $\mathbf{v}$ and (8) tested against every z . I claim this is equivalent to testing their sum against every pair $(\mathbf{v}, z)$.

One direction is immediate: if (7) and (8) both hold then so does their sum, for every pair. The other direction is the one that needs an argument. Suppose the summed equation holds for all $(\mathbf{v}, z) \in \mathbf{V}_{\text{div}} \times W$. Choosing $z = 0$ makes every term coming from (8) vanish, and what is left is exactly (7) for arbitrary $\mathbf{v}$. Choosing $\mathbf{v} = \mathbf{0}$ recovers (8) in the same way. So both original equations follow from the sum, and the two formulations carry exactly the same information. Nothing is lost by adding them, and it is the freedom to set one component of the test pair to zero that makes the step reversible.

Adding (7) and (8) therefore gives the weak problem in its final form: find $(\mathbf{u}, w) \in \mathbf{V}_{\text{div}} \times W$ such that

$$a((\mathbf{u}, w), (\mathbf{v}, z)) = F(\mathbf{v}, z) \quad \text{for all } (\mathbf{v}, z) \in \mathbf{V}_{\text{div}} \times W, \quad (9)$$

where

$$\begin{aligned} a((\mathbf{u}, w), (\mathbf{v}, z)) &= (\nu + \nu_r)(\nabla \mathbf{u}, \nabla \mathbf{v}) + (c_a + c_d)(\nabla w, \nabla z) + 4\nu_r(w, z) \\ &\quad - 2\nu_r(\nabla^\perp w, \mathbf{v}) - 2\nu_r(\text{curl } \mathbf{u}, z) \end{aligned} \quad (10)$$

and $F(\mathbf{v}, z) = (\mathbf{f}, \mathbf{v}) + (g, z)$.

The first three terms are things we have already met. The first is the Stokes term. The second and third together are exactly the reaction diffusion problem from Lesson 1, Problem C, with $c = c_a + c_d$ and $\kappa = 4\nu_r$. The last two are the couplings, and they are the new difficulty, because they carry a minus sign and there is nothing forcing them to be small. They can be negative, and if they are large and negative they could in principle destroy coercivity.

Remark 2.1 (Why we bother adding the equations). It would be perfectly possible to keep (7) and (8) as two separate equations, and nothing would be wrong with that. The reason for combining them is that Lax–Milgram, Lemma 1.3, is a statement about one bilinear form on one Hilbert space. It says nothing about a pair of coupled equations posed on two different spaces. By treating the pair $(\mathbf{u}, w)$ as a single unknown living in the single product space $\mathbf{V}_{\text{div}} \times W$, every result from Lesson 1 becomes applicable without any structural change, and indeed the coercivity proof below turns out to be the Lesson 1 argument with one extra absorption step.

This packaging is the standard way of handling coupled systems and it is worth having as a reflex. When faced with a system of several equations, look for the product space that turns it into one.

3 Coercivity

3.1 Plan of the argument, and why ε comes first

Before doing the calculation I want to say what the shape of it is, because the order of operations is not obvious the first time.

We set $\mathbf{v} = \mathbf{u}$ and $z = w$, which turns the first three terms of (10) into norms immediately and leaves one term of unknown sign. That bad term is a cross term: it is an inner product of two different quantities, $\text{curl } \mathbf{u}$ and w , and there is no reason for it to be small. So we bound it, and the only way to bound a cross term in terms of squares is Cauchy–Schwarz followed by Young.

The point of Remark 1.5 is that Young leaves us a free parameter, and until we fix it we do not know how much damage has been done to each of the good terms. So the parameter has to be chosen, and it has to be chosen at this stage rather than later, because the whole question of whether coercivity holds is the question of whether there exists a choice of ε making all the remaining coefficients non-negative. If no such ε exists, the argument fails and we would have to conclude something different about the system. So finding ε is not a technical detail postponed to the end. It is the crux.

Once ε is fixed we substitute it in, the constants become explicit numbers, and after that the parameter never appears again. This is why ε is used in the coercivity proof and does not appear anywhere in the boundedness proof. In boundedness there is nothing to absorb, because the two arguments $(\mathbf{u}, w)$ and $(\mathbf{v}, z)$ are independent and there is no possibility of one term cancelling another. Every term simply gets bounded on its own and the constants are added up.

3.2 Step 1: setting the arguments equal

Put $\mathbf{v} = \mathbf{u}$ and $z = w$ in (10). The first three terms become norms straight away, since $(\phi, \phi) = \|\phi\|_{L^2}^2$ directly from the definition of the norm. There is no estimate at this step:

$$\begin{aligned} (\nu + \nu_r) (\nabla \mathbf{u}, \nabla \mathbf{u}) &= (\nu + \nu_r) \|\nabla \mathbf{u}\|_{L^2}^2, \\ (c_a + c_d) (\nabla w, \nabla w) &= (c_a + c_d) \|\nabla w\|_{L^2}^2, \\ 4\nu_r (w, w) &= 4\nu_r \|w\|_{L^2}^2. \end{aligned}$$

For the fourth term we use Lemma 1.6 with $\mathbf{v} = \mathbf{u}$, together with the symmetry of the L^2 inner product:

$$(\nabla^\perp w, \mathbf{u}) = (w, \text{curl } \mathbf{u}) = (\text{curl } \mathbf{u}, w).$$

So the fourth term becomes $-2\nu_r (\text{curl } \mathbf{u}, w)$, which is identical to the fifth term. The two couplings therefore combine into a single term:

$$-2\nu_r (\text{curl } \mathbf{u}, w) - 2\nu_r (\text{curl } \mathbf{u}, w) = -4\nu_r (\text{curl } \mathbf{u}, w).$$

Collecting everything,

$$a((\mathbf{u}, w), (\mathbf{u}, w)) = (\nu + \nu_r) \|\nabla \mathbf{u}\|_{L^2}^2 + (c_a + c_d) \|\nabla w\|_{L^2}^2 + 4\nu_r \|w\|_{L^2}^2 - 4\nu_r (\text{curl } \mathbf{u}, w). \quad (11)$$

This is an identity, not an estimate. Three positive terms and one whose sign we do not know.

3.3 Step 2: bounding the coupling term

Apply Cauchy–Schwarz (Lemma 1.1) to the last term of (11):

$$|4\nu_r (\text{curl } \mathbf{u}, w)| \leq 4\nu_r \|\text{curl } \mathbf{u}\|_{L^2} \|w\|_{L^2}. \quad (12)$$

We now have a product of two different norms, and nothing in (11) to compare it against directly, since everything there is squared. So apply Young (Lemma 1.4) with

$$a = \|\operatorname{curl} \mathbf{u}\|_{L^2}, \quad b = \|w\|_{L^2},$$

and a parameter $\varepsilon > 0$ still to be chosen:

$$\|\operatorname{curl} \mathbf{u}\|_{L^2} \|w\|_{L^2} \leq \frac{\varepsilon}{2} \|\operatorname{curl} \mathbf{u}\|_{L^2}^2 + \frac{1}{2\varepsilon} \|w\|_{L^2}^2. \quad (13)$$

The first term on the right involves $\|\operatorname{curl} \mathbf{u}\|_{L^2}^2$, which does not appear anywhere in (11). What does appear is $\|\nabla \mathbf{u}\|_{L^2}^2$. Lemma 1.7 is the bridge between them:

$$\|\operatorname{curl} \mathbf{u}\|_{L^2}^2 \leq \|\nabla \mathbf{u}\|_{L^2}^2. \quad (14)$$

Note that on $\mathbf{V}_{\operatorname{div}}$ we actually have equality here, because $\operatorname{div} \mathbf{u} = 0$, so nothing is being wasted. I write it as an inequality because that is the form that survives if the divergence free constraint is later relaxed.

Combining (12), (13) and (14),

$$|4\nu_r (\operatorname{curl} \mathbf{u}, w)| \leq 4\nu_r \left[\frac{\varepsilon}{2} \|\nabla \mathbf{u}\|_{L^2}^2 + \frac{1}{2\varepsilon} \|w\|_{L^2}^2 \right] = 2\nu_r \varepsilon \|\nabla \mathbf{u}\|_{L^2}^2 + \frac{2\nu_r}{\varepsilon} \|w\|_{L^2}^2. \quad (15)$$

3.4 Step 3: substituting back

Since $-X \geq -|X|$ for any real X , we may replace the coupling term in (11) by minus the bound (15):

$$\begin{aligned} a((\mathbf{u}, w), (\mathbf{u}, w)) &\geq (\nu + \nu_r) \|\nabla \mathbf{u}\|_{L^2}^2 + (c_a + c_d) \|\nabla w\|_{L^2}^2 + 4\nu_r \|w\|_{L^2}^2 \\ &\quad - 2\nu_r \varepsilon \|\nabla \mathbf{u}\|_{L^2}^2 - \frac{2\nu_r}{\varepsilon} \|w\|_{L^2}^2 \\ &= \left(\nu + \nu_r - 2\nu_r \varepsilon \right) \|\nabla \mathbf{u}\|_{L^2}^2 + (c_a + c_d) \|\nabla w\|_{L^2}^2 + \left(4\nu_r - \frac{2\nu_r}{\varepsilon} \right) \|w\|_{L^2}^2. \end{aligned} \quad (16)$$

3.5 Step 4: choosing ε

Now we choose the parameter. There are two things we want.

First, we want the coefficient of $\|\nabla \mathbf{u}\|_{L^2}^2$ to be free of ν_r , that is, we want the ν_r contributions to cancel:

$$\nu_r - 2\nu_r \varepsilon = 0. \quad (17)$$

Second, we need the coefficient of $\|w\|_{L^2}^2$ to be non-negative, since a negative coefficient there would leave a term working against us:

$$4\nu_r - \frac{2\nu_r}{\varepsilon} \geq 0. \quad (18)$$

That is two conditions and one unknown, so it is not obvious in advance that both can be met. Solve (17). Dividing by $\nu_r > 0$,

$$1 - 2\varepsilon = 0 \quad \implies \quad \varepsilon = \frac{1}{2}.$$

Now check (18) at this value. With $\varepsilon = 1/2$ we have $2\nu_r/\varepsilon = 4\nu_r$, so

$$4\nu_r - \frac{2\nu_r}{\varepsilon} = 4\nu_r - 4\nu_r = 0 \geq 0,$$

which is satisfied, though only just: the coefficient is exactly zero, not positive.

Remark 3.1 (This is not a coincidence). The one value of ε that removes ν_r from the velocity coefficient is the same value that exactly exhausts the damping term $4\nu_r \|w\|_{L^2}^2$. Both constraints bind at once, with nothing to spare.

This happens because the coefficients in the micropolar model are not independent. The two couplings carry $2\nu_r$ each and the damping term carries $4\nu_r$, and these are tied together by the way the model is derived from conservation of angular momentum. The algebra above is the mathematical shadow of that physical relationship. I say more about this in Section 6.

Substituting $\varepsilon = 1/2$ into (16), the velocity coefficient becomes $\nu + \nu_r - \nu_r = \nu$ and the $\|w\|_{L^2}^2$ term disappears entirely:

$$\boxed{a((\mathbf{u}, w), (\mathbf{u}, w)) \geq \nu \|\nabla \mathbf{u}\|_{L^2}^2 + (c_a + c_d) \|\nabla w\|_{L^2}^2.} \quad (19)$$

Two things are worth noticing. The coefficient of the velocity term is ν , not $\nu + \nu_r$: the vortex viscosity contribution was entirely consumed in absorbing the coupling. And ν_r does not appear anywhere on the right hand side.

3.6 Step 5: converting to the full product norm

Estimate (19) controls only the gradient parts. The norm (6) also contains $\|\mathbf{u}\|_{L^2}^2$ and $\|w\|_{L^2}^2$, and we have neither, since the $\|w\|_{L^2}^2$ term was used up. Both fields have zero trace, so Poincaré supplies what is missing. This is the Lesson 1 argument run twice.

By Lemma 1.2,

$$\|\mathbf{u}\|_{L^2}^2 \leq C_\Omega^2 \|\nabla \mathbf{u}\|_{L^2}^2, \quad \|w\|_{L^2}^2 \leq C_\Omega^2 \|\nabla w\|_{L^2}^2,$$

so

$$\|\mathbf{u}\|_{H^1}^2 = \|\mathbf{u}\|_{L^2}^2 + \|\nabla \mathbf{u}\|_{L^2}^2 \leq (C_\Omega^2 + 1) \|\nabla \mathbf{u}\|_{L^2}^2,$$

and identically for w . Rearranging both,

$$\|\nabla \mathbf{u}\|_{L^2}^2 \geq \frac{\|\mathbf{u}\|_{H^1}^2}{C_\Omega^2 + 1}, \quad \|\nabla w\|_{L^2}^2 \geq \frac{\|w\|_{H^1}^2}{C_\Omega^2 + 1}. \quad (20)$$

Substituting (20) into (19),

$$a((\mathbf{u}, w), (\mathbf{u}, w)) \geq \frac{\nu}{C_\Omega^2 + 1} \|\mathbf{u}\|_{H^1}^2 + \frac{c_a + c_d}{C_\Omega^2 + 1} \|w\|_{H^1}^2.$$

Finally, replace both coefficients by the smaller of the two, exactly as in Lesson 1 Problem C:

$$a((\mathbf{u}, w), (\mathbf{u}, w)) \geq \frac{\min(\nu, c_a + c_d)}{C_\Omega^2 + 1} (\|\mathbf{u}\|_{H^1}^2 + \|w\|_{H^1}^2).$$

We have proved the following.

Theorem 3.2 (Coercivity). *For all $(\mathbf{u}, w) \in \mathbf{V}_{\text{div}} \times W$,*

$$a((\mathbf{u}, w), (\mathbf{u}, w)) \geq \alpha \|(\mathbf{u}, w)\|^2, \quad \alpha = \frac{\min(\nu, c_a + c_d)}{C_\Omega^2 + 1}.$$

In particular α does not depend on the vortex viscosity ν_r .

4 Boundedness

4.1 Why there is no ε here

In the coercivity proof the two arguments were the same, which is what allowed one term to be absorbed into another. In boundedness the two arguments $(\mathbf{u}, w)$ and $(\mathbf{v}, z)$ are completely independent, so no cancellation between terms is possible and there is nothing to absorb. Each term is bounded separately by Cauchy–Schwarz and the constants are added. Young’s inequality plays no role.

Throughout we use, repeatedly, the fact that each individual piece is bounded by the full product norm. From (6), all four summands are non-negative, so discarding any of them gives, for instance,

$$\|\nabla \mathbf{u}\|_{L^2} \leq \|\mathbf{u}\|_{H^1} \leq \|(\mathbf{u}, w)\|, \quad \|w\|_{L^2} \leq \|w\|_{H^1} \leq \|(\mathbf{u}, w)\|, \quad (21)$$

and similarly for every other piece. This is the same discard-a-non-negative-term move as before.

4.2 The five terms

Term 1. By Cauchy–Schwarz and then (21),

$$(\nu + \nu_r) |(\nabla \mathbf{u}, \nabla \mathbf{v})| \leq (\nu + \nu_r) \|\nabla \mathbf{u}\|_{L^2} \|\nabla \mathbf{v}\|_{L^2} \leq (\nu + \nu_r) \|(\mathbf{u}, w)\| \|(\mathbf{v}, z)\|.$$

Term 2. Identically,

$$(c_a + c_d) |(\nabla w, \nabla z)| \leq (c_a + c_d) \|\nabla w\|_{L^2} \|\nabla z\|_{L^2} \leq (c_a + c_d) \|(\mathbf{u}, w)\| \|(\mathbf{v}, z)\|.$$

Term 3. Again,

$$4\nu_r |(w, z)| \leq 4\nu_r \|w\|_{L^2} \|z\|_{L^2} \leq 4\nu_r \|(\mathbf{u}, w)\| \|(\mathbf{v}, z)\|.$$

Term 4. Here we first use Lemma 1.6 to move the derivative onto $\mathbf{v}$, then Cauchy–Schwarz, then Lemma 1.7 to replace the curl by the full gradient:

$$2\nu_r \left| \left(\nabla^\perp w, \mathbf{v} \right) \right| = 2\nu_r |(w, \operatorname{curl} \mathbf{v})| \leq 2\nu_r \|w\|_{L^2} \|\operatorname{curl} \mathbf{v}\|_{L^2} \leq 2\nu_r \|w\|_{L^2} \|\nabla \mathbf{v}\|_{L^2},$$

and then by (21) this is at most $2\nu_r \|(\mathbf{u}, w)\| \|(\mathbf{v}, z)\|$.

Term 5. The same, with the roles of the two arguments exchanged:

$$2\nu_r |(\operatorname{curl} \mathbf{u}, z)| \leq 2\nu_r \|\operatorname{curl} \mathbf{u}\|_{L^2} \|z\|_{L^2} \leq 2\nu_r \|\nabla \mathbf{u}\|_{L^2} \|z\|_{L^2} \leq 2\nu_r \|(\mathbf{u}, w)\| \|(\mathbf{v}, z)\|.$$

4.3 Adding up

By the triangle inequality applied to (10), the modulus of the whole form is at most the sum of the five bounds. Collecting the constants,

$$M = (\nu + \nu_r) + (c_a + c_d) + 4\nu_r + 2\nu_r + 2\nu_r = \nu + (c_a + c_d) + 9\nu_r.$$

Theorem 4.1 (Boundedness). *For all $(\mathbf{u}, w), (\mathbf{v}, z) \in \mathbf{V}_{\operatorname{div}} \times W$,*

$$|a((\mathbf{u}, w), (\mathbf{v}, z))| \leq M \|(\mathbf{u}, w)\| \|(\mathbf{v}, z)\|, \quad M = \nu + (c_a + c_d) + 9\nu_r.$$

In particular M grows linearly in ν_r .

5 Well-posedness

Theorem 5.1 (Well-posedness). *Let $\nu > 0$, $c_a + c_d > 0$, $\nu_r \geq 0$, and let $\mathbf{f} \in [L^2(\Omega)]^2$ and $g \in L^2(\Omega)$. Then the weak problem (9) has exactly one solution $(\mathbf{u}, w) \in \mathbf{V}_{\text{div}} \times W$, and it satisfies*

$$\|(\mathbf{u}, w)\| \leq \frac{1}{\alpha} \|F\|, \quad \alpha = \frac{\min(\nu, c_a + c_d)}{C_\Omega^2 + 1}.$$

Proof. The product space $\mathbf{V}_{\text{div}} \times W$ is a Hilbert space with the norm (6), being a closed subspace of a product of Hilbert spaces. The form a is bilinear by inspection, bounded by Theorem 4.1, and coercive by Theorem 3.2. The functional F is bounded because, by Cauchy–Schwarz,

$$|F(\mathbf{v}, z)| \leq \|\mathbf{f}\|_{L^2} \|\mathbf{v}\|_{L^2} + \|g\|_{L^2} \|z\|_{L^2} \leq (\|\mathbf{f}\|_{L^2} + \|g\|_{L^2}) \|(\mathbf{v}, z)\|.$$

Lemma 1.3 then gives existence, uniqueness, and the stated bound. $\square$

6 Conclusions

6.1 Summary of the constants

Quantity	Value	Depends on ν_r ?
Coercivity constant α	$\frac{\min(\nu, c_a + c_d)}{C_\Omega^2 + 1}$	no
Boundedness constant M	$\nu + (c_a + c_d) + 9\nu_r$	yes, linearly
Ratio M/α	$\frac{(\nu + (c_a + c_d) + 9\nu_r)(C_\Omega^2 + 1)}{\min(\nu, c_a + c_d)}$	yes, linearly

6.2 The limit of small vortex viscosity

As $\nu_r \rightarrow 0$, both couplings and the damping term all vanish, since each carries a factor ν_r . The bilinear form reduces to

$$a((\mathbf{u}, w), (\mathbf{v}, z)) \longrightarrow \nu (\nabla \mathbf{u}, \nabla \mathbf{v}) + (c_a + c_d) (\nabla w, \nabla z),$$

which is completely decoupled. The velocity solves a divergence free Stokes problem and the microrotation solves a Poisson problem, and neither knows about the other. This makes sense physically: ν_r is the coupling. With no vortex viscosity the microstructure spins freely and exchanges nothing with the bulk flow.

The coercivity constant survives this limit unchanged, and each of the two constants inside the minimum is precisely the Lesson 1 constant for the corresponding decoupled problem. So the limit is continuous, and the analysis degenerates to the right thing, which is a good check that no error has crept in.

6.3 The limit of large vortex viscosity, and what it predicts numerically

As $\nu_r \rightarrow \infty$ the situation is different. Coercivity never fails: α is fixed and strictly positive for every ν_r , so by Theorem 5.1 the problem remains uniquely solvable no matter how large the vortex viscosity is. But M grows without bound, so the ratio

$$\frac{M}{\alpha} \sim \frac{9\nu_r (C_\Omega^2 + 1)}{\min(\nu, c_a + c_d)} \quad \text{as } \nu_r \rightarrow \infty$$

grows linearly. This ratio is the condition number of the problem, and it is exactly the constant that appears in Céa's lemma:

$$\|(\mathbf{u}, w) - (\mathbf{u}_h, w_h)\| \leq \frac{M}{\alpha} \inf_{(\mathbf{v}_h, z_h)} \|(\mathbf{u}, w) - (\mathbf{v}_h, z_h)\|.$$

So the analysis makes a concrete numerical prediction. The convergence *rates* of a finite element discretisation should be unaffected by ν_r , because they come from the interpolation estimate on the right hand side. But the error *constants* should grow linearly in ν_r , meaning that for a fixed mesh the computed solution gets worse as the vortex viscosity increases, and a finer mesh is needed to recover the same accuracy. The discrete linear system should also become progressively harder for iterative solvers, so Krylov iteration counts should climb and preconditioning should stop being optional.

This is a prediction that can be tested directly, and testing it is what the parameter study in the project plan is for. Having derived the prediction from the analysis first, rather than simply running the code and describing what came out, is the difference between numerical analysis and computation.

6.4 Why ν_r disappears from the coercivity constant

The vortex viscosity appears four times in the equations, and yet it leaves no trace at all in α . I think this is the most interesting thing in these notes, and the reason for it is physical rather than algebraic.

Coercivity is fundamentally an energy statement. It says that the bilinear form controls the total energy of the pair $(\mathbf{u}, w)$ from below, which is the same as saying that the system dissipates energy at a rate proportional to how much energy it has. Now consider what ν_r actually does. The two coupling terms transfer angular momentum between the bulk flow and the microstructure, one in each direction. Transfer is not dissipation: energy moved from the velocity field to the microrotation field is still in the system. A mechanism that only moves energy around internally cannot contribute anything to the total, so it must cancel out of any statement about the total, and that is exactly what the choice $\varepsilon = 1/2$ achieves.

The energy that genuinely leaves the system is dissipated by the two diffusivities: ν , through viscous shear in the bulk fluid, and $c_a + c_d$, through angular diffusion in the microstructure. These are the only two mechanisms that actually destroy energy, and so these are the only two that appear in α . The minimum appears because the system as a whole can only dissipate as fast as its slowest channel.

This also explains Remark 3.1. The two conditions (17) and (18) bind simultaneously because the coefficients $2\nu_r$, $2\nu_r$ and $4\nu_r$ in the model are not independent parameters. They come from a single conservation law, and the exact cancellation in the algebra is the trace that conservation law leaves behind. If the model had been written with arbitrary unrelated coefficients, the two conditions would generally have been incompatible and coercivity would have required a smallness assumption on the coupling. The fact that it does not is a statement about the model, not about the proof.

6.5 What comes next

The result proved here is genuine but incomplete, because the space $\mathbf{V}_{\text{div}}$ cannot be discretised conveniently. Constructing a finite element basis of divergence free functions is possible but awkward, and essentially nobody does it. The practical route is to keep the pressure as an unknown, work on $[H_0^1(\Omega)]^2 \times L_0^2(\Omega) \times H_0^1(\Omega)$, and impose $\text{div } \mathbf{u} = 0$ weakly through the pressure.

The price is that coercivity is no longer enough. The pressure enters through a constraint rather than through a coercive term, so Lax–Milgram does not apply and must be replaced by

the Babuška–Brezzi theory, whose extra hypothesis is the inf-sup condition. That is the subject of Lesson 3, and it is the real reason mixed finite element methods exist as a separate subject.

Nothing proved here is wasted. The coercivity estimate (19) is exactly the first of the two Brezzi conditions, and it carries over unchanged. What has to be added is the second.